Why APL feedback makes the effect of blocking uneven across odors

> **QUESTION**
> Why does blocking a uniform, feedback APL raise the similarity of some odor pairs more than others, and do the recordings show the same signature?

INPUTS

- The pinned model of Report 07 (uniform APL, hemibrain R and male CNS L and R)
- The recordings (Report 06)

OUTPUTS

- `odor_weighted/TNT_feedback_vs_fixed.csv`, `TNT_feedback_vs_fixed.json`
- `odor_weighted/chance_overlap_simulation.png`, `.csv`, `.json`, `chance_overlap_data_pairs.csv`

SCRIPTS

- `scripts/tnt_feedback_vs_fixed.py`
- `scripts/chance_overlap_sim.py`
- `scripts/reports/r08_feedback_uneven.py` (this report)

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Report 07 found that a uniform APL with feedback predicts which pairs gain similarity when APL is blocked about as well as any wiring version, and better than an inhibition of fixed size. This report explains why, in two steps. First, feedback makes the inhibition odor-dependent, so blocking it adds the most responders to strong odors. Second, an odor that gains responders becomes more similar to every other odor by chance overlap alone.

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range
ϕ_o	Fraction of KCs (model) or units (data) responding to odor o	0 to 1
γ_o	Gain of odor o when APL is blocked: $\phi_o^{\text{blocked}}/\phi_o^{\text{intact}}$ (in the data, TNT ÷ TNT3c; called s_o in Reports 06 and 07)	×
CV	Coefficient of variation across odors: standard deviation ÷ mean	dimensionless
$n_a,\ n_{ab}$	Number of units responding to odor a, and to both a and b	counts
N	Number of units (or KCs)	
p_a	n_a/N, the fraction responding to a (same as ϕ_a)	0 to 1
$\sqrt{p_a p_b}$	Chance overlap: expected cosine of two unrelated 0/1 codes	0 to 1
π_k	Probability that unit k responds to a random odor (its excitability)	0 to 1
ΔS_{ab}	Change in similarity when APL is blocked	difference in cosine

1 Step 1: feedback inhibits strong odors more

With a uniform APL, every KC gets the same inhibition aA_o for odor o (Report 05). It is the same for every KC, but not for every odor: A_o is the mean KC response, so an odor that drives many KCs recruits a lot of APL and is held down hard, and one that drives few recruits little. Feedback therefore evens out how many KCs the odors activate, like a normalization, and blocking APL undoes it. To show that feedback is what matters, three other rules for the inhibition are compared, all with the same blocked responses and the same mean intact fraction (TNT3c's):

- **feedback**: uniform APL feedback (version U of Report 07);
- **fixed**: one threshold raise, the same for every odor, set so the mean fraction matches TNT3c;
- **same share**: a raise per odor so that every odor keeps the same share of its blocked responders;
- **TNT3c per odor**: a raise per odor so that each odor's intact fraction equals its own TNT3c fraction (uses the data's per-odor answer).

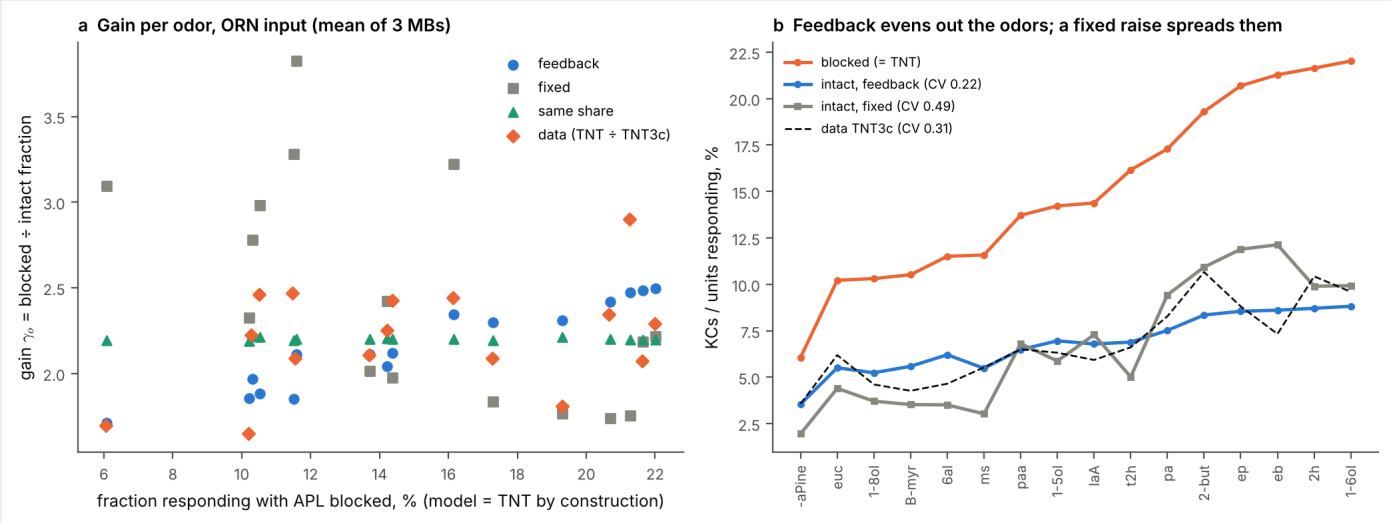


Figure 1. Feedback acts like a normalization across odors. a, Each point is one odor. Horizontal: the fraction responding with APL blocked (the model is pinned to TNT). Vertical: how many times more respond blocked than intact. Blue: feedback. Grey: fixed raise. Green: same share. Orange diamonds: data. **b**, Per odor (sorted by the blocked fraction), the percentage responding blocked (orange), intact with feedback (blue) and intact with a fixed raise (grey), with the data's TNT3c values (dashed). CV is the spread across odors.

With feedback, the gain rises with odor strength (Spearman +0.96 over odors), from ×1.71 to ×2.50. A fixed raise does the opposite (Spearman -0.62): weak odors lose the largest share of their few responders, because a fixed raise removes a fixed slice of the drive distribution near threshold.

2 Step 2: from more responders to more similarity

Denser codes overlap more by chance. For 0/1 responses the cosine similarity is the overlap divided by the geometric mean of the code sizes:

$$\cos(a, b) = \frac{n_{ab}}{\sqrt{n_a\, n_b}} \tag{1}$$

If odor b's responders are placed at random, independently of a's, the overlap n_{ab} follows a hypergeometric distribution with mean $n_a n_b / N$. The expected cosine is therefore

$$\mathbb{E}[\cos(a,b)] = \frac{n_a n_b / N}{\sqrt{n_a n_b}} = \sqrt{p_a\, p_b}, \qquad \mathrm{SD} \approx \sqrt{\frac{(1-p_a)(1-p_b)}{N-1}} \tag{2}$$

So an odor that gains responders becomes more similar to every other odor, and its whole row of the similarity matrix rises. This is the link between Step 1 (per-odor gains) and the per-odor part of ΔS (Report 06, Section 4).

A floor, not the exact chance level, for KCs

The formula assumes every unit is equally likely to respond. If unit k responds with probability π_k, the expected cosine is about $(p_a p_b + \mathrm{cov}_k(\pi_a, \pi_b))/\sqrt{p_a p_b}$, which is higher when the same units respond easily to both odors. In the connectome model, two unrelated odors that each drive 10% of KCs have a cosine of 1.59 × $\sqrt{p_a p_b}$ with the ORN-table glomeruli, and 1.18 × with every glomerulus responding, because some KCs are easier to drive than others.

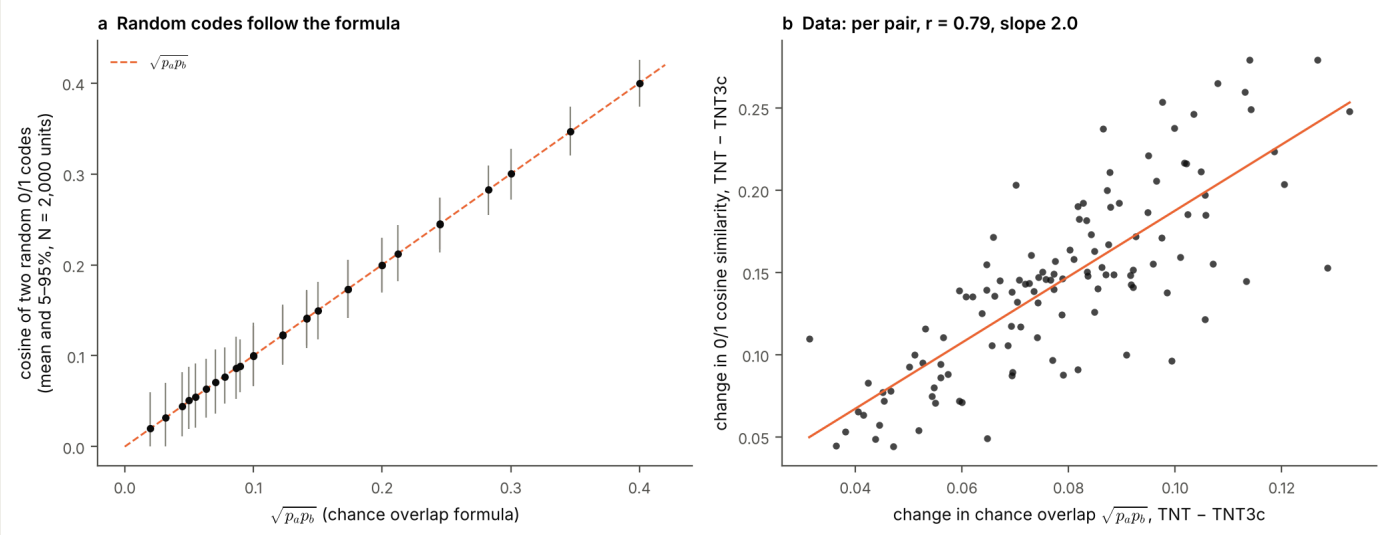


Figure 2. Chance overlap and the similarity change. a, Simulation: for pairs of random, unrelated 0/1 codes over 2,000 units, the mean cosine (dots) and its 5–95% range (grey) against $\sqrt{p_a p_b}$; the dashed line is equality. **b**, Data: each point is an odor pair. Horizontal: the change in $\sqrt{p_a p_b}$ between TNT and TNT3c, computed per recording from the fractions responding (> 1) and averaged. Vertical: the change in cosine similarity of the 0/1 responses. The orange line is the least-squares fit.

In the data the real 0/1 similarity is about twice $\sqrt{p_a p_b}$ in both conditions (0.145 vs 0.066 in TNT3c, 0.289 vs 0.144 in TNT). That excess can come from related odors or from units that respond to many odors. A shuffle that keeps each unit's number of responses and each odor's fraction reproduces the mean (0.145 in TNT3c), so with 16 odors the two cannot be told apart.

3 Step 3: which inhibition rule fits the data

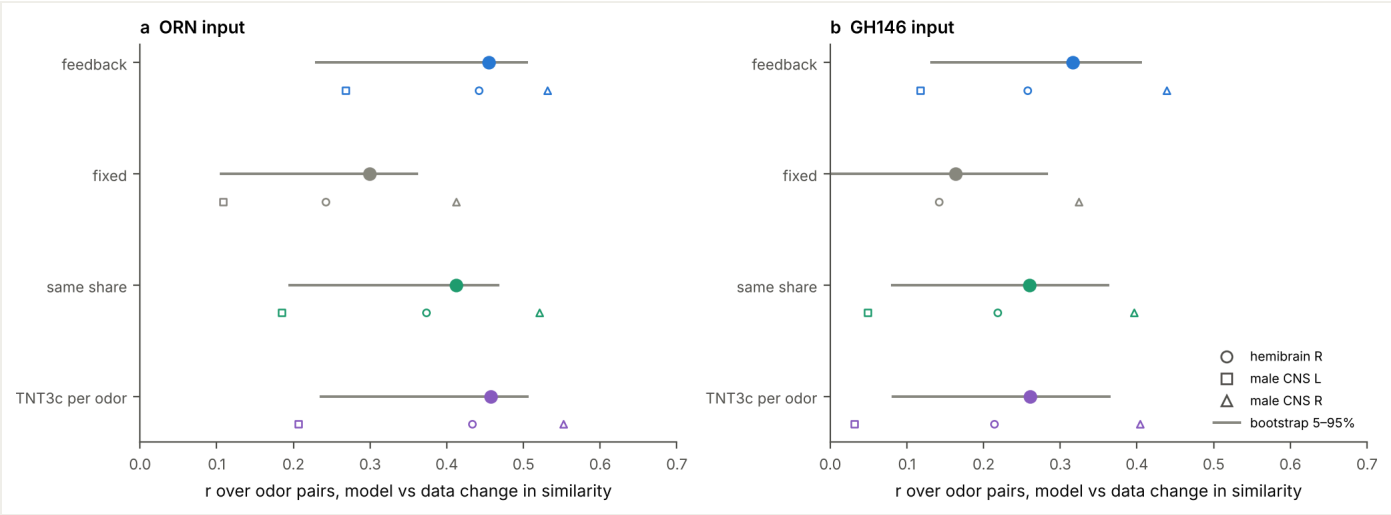


Figure 3. Fit of the four inhibition rules. For each inhibition rule, the large dot is r between the model's change in similarity (mean of the three mushroom bodies) and the data's, over odor pairs. The grey line is its bootstrap 5–95% range over recordings. The small open markers are the single mushroom bodies.

- Feedback fits better than a fixed raise in every mushroom body and in 100% of bootstrap draws (r 0.46 vs 0.30 with ORN input).
- Feedback does as well as setting each odor's intact fraction from the data (r 0.46), although it is not given the per-odor TNT3c fractions.
- The per-odor part of the change matches the data with Spearman 0.52 for feedback and 0.20 for a fixed raise.

4 Step 4: is the feedback signature in the recordings?

If APL acted as uniform feedback, odors that drive more units should gain more units when APL is blocked. To test this without shared noise, odor strength and gain are measured on separate halves of the recordings: 4 TNT and 5 TNT3c recordings give the strength (fraction responding, averaged over the two conditions), and the other 4 TNT and 6 TNT3c give the gain. This is repeated for 2,000 random splits.

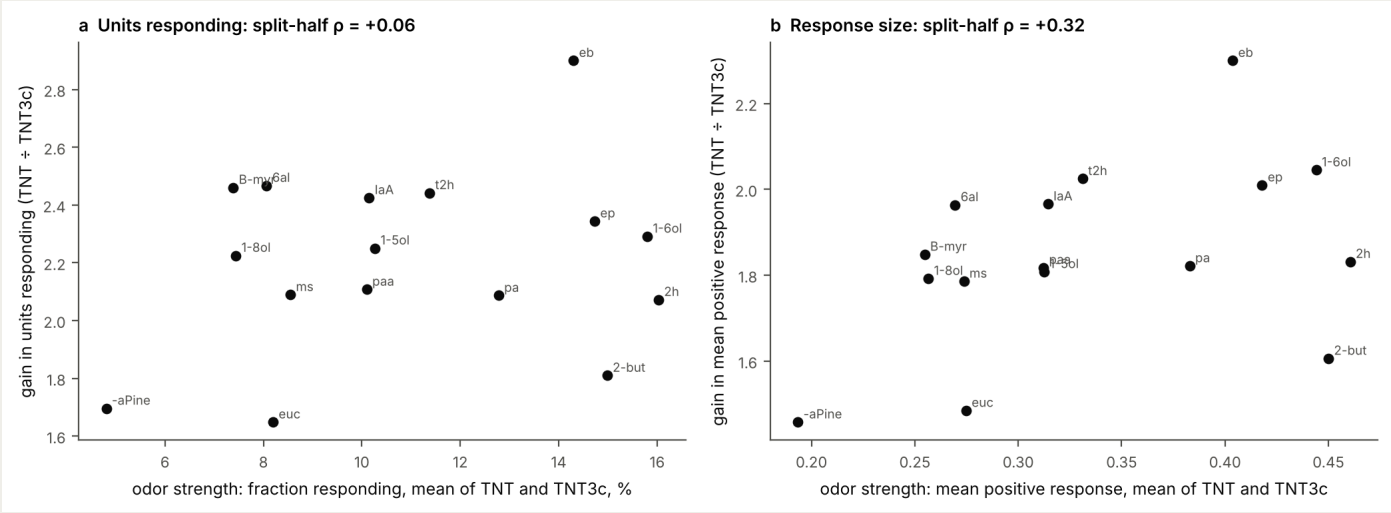


Figure 4. The feedback signature in the data. Each point is one odor, using all recordings (the split-half Spearman ρ in the titles uses separate halves for the two axes, mean of 2,000 splits). **a**, Odor strength as the fraction of units responding (> 1) against how many times more units respond in TNT. **b**, Odor strength as the mean positive response (negative responses set to 0, then averaged over all units; response units as provided) against its ratio TNT ÷ TNT3c.

- **Units responding**: the gain is unrelated to odor strength (mean split-half Spearman +0.06, 5–95% -0.22 to +0.33). The spread across odors is the same in TNT and TNT3c (CV 0.32 and 0.31), whereas uniform feedback predicts less spread with APL intact.
- **Response size**: stronger odors do gain more (Spearman +0.32 for the mean positive response, positive in 99% of splits; +0.37 for the root mean square response).
- So the recordings show the feedback signature in response size but not in the number of responding units.

Caveats

The rules other than feedback are not mechanisms; they are reference points with the same mean inhibition.

Noise inflates TNT3c's apparent number of weak responders differently from TNT's, which can blur the gain-vs-strength relation.

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